Inscribed angles and polygons

1

a $x = 66$

b $x = 76$

c $x = 41$

d $x = 4$

e $x = 12$

f $m = 101^\circ$

g $x = 108^\circ, y = 60^\circ$

h $y = 60^\circ$

i $m = 36^\circ$

2 150°

3 $m\angle ADB = 56^\circ$

4 $m\angle RPQ = 73^\circ$

5 $m\angle ACB = 90^\circ$

6

a $p = 90^\circ$

b $q = 63^\circ$

c $r = 117^\circ$

Additional questions

7

a $m\angle CDB = 32^\circ$

b $m\overset{\frown}{RS} = 70^\circ$

c $m\angle BAC = 40^\circ$

d $m\angle MPN = 35^\circ$

e $m\overset{\frown}{GH} = 122^\circ$

f $m\angle XFD = 57^\circ$

8

a $m\overset{\frown}{DC} = 122^\circ$

b $m\angle Q = 105^\circ$

9

a $m\angle P = 124^\circ$

10

a $m\angle UPT = 30^\circ$

b $m\overset{\frown}{US} = 60^\circ$

11 $x = 3$

12

a $x = 20$

b $m\overset{\frown}{SV} = 110^\circ$

13 $m\angle F = 180^\circ - a^\circ$

14

a $m\angle 1 = 60^\circ$

b $m\overset{\frown}{BD} = 168^\circ$

15

a $x = \frac{40}{9}$

b $m\overset{\frown}{GF} = 112^\circ$

16 The angles are congruent because they are all inscribed angles with the same intercepted arc

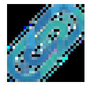
17 $\angle CAD$ and $\angle CBD$ have the same intercepted arc, $\overset{\frown}{CD}$. Since $\angle CAD$ is a central angle, $m\overset{\frown}{CD} = m\angle CAD$. Since $\angle CBD$ is an inscribed angle, $m\overset{\frown}{CD} = 2(m\angle CBD)$. Using substitution we get $m\angle CAD = 2(m\angle CBD)$, therefore $y = 2x$.

18

a 240°

b $\overset{\frown}{FEA}$, $\overset{\frown}{CDB}$, and $\overset{\frown}{BFE}$

c $m\overset{\frown}{COF} = \frac{1}{2}m\overset{\frown}{CAF}$



19

a 45°

b 90°

c $\frac{1}{2}\left(\frac{32-2n}{32} * 360\right)^{\circ}$